

SHARP INVERSE GROWTH, SINGULAR DEFECT, AND FINITE HEAT LIMITS FOR THE LITERAL ERDŐS–STRAUS RECEIVER

THE CLANKERS

ABSTRACT. For every existing positive integral Erdős–Straus witness at a prime $p \equiv 1 \pmod{12}$, we prove sharp prime-growth bounds for an explicit four-mark complex receiving map. In its original ordered label and receiving coordinates,

$$\|O^{-1}\|_2 = O(p^{3/2}), \quad \|\wedge^2 O^{-1}\|_2 = O(1), \quad \|\wedge^3 O^{-1}\|_2 = O(p^{-2}).$$

Only one inverse singular direction can become unbounded. Two infinite prime families in the class $13 \pmod{24}$ attain all three exponents; their complete singular constants, a rank-one inverse limit, and a finite receiver heat limit are evaluated. Separately, extend the same receiver to the ambient space of four distinct positive labelled roots without imposing the reciprocal identity. On its exact exceptional divisor the receiver has rank three, with an explicit kernel and cokernel hyperplane; the global sign theorem proves that this divisor does not meet the positive integral witness locus. A fixed- c integral family has four computed first singular corrections and an exhaustive observation hierarchy with powers $0, 2, 3, 15$. The proof retains the two exact arithmetic channels, every root label and square-root sign, the standard Hermitian coordinates, and the conductor and metric factors. Three exact positive-coefficient comparisons and a high-precision replay accompany the proof. The arithmetic receiver bounds begin with an existing witness and are not an existence theorem for the Erdős–Straus equation.

1. LITERAL RECEIVER AND ARITHMETIC DOMAIN

Let $p \equiv 1 \pmod{12}$ be prime and let

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{Y} + \frac{1}{Z}, \quad a, Y, Z \in \mathbb{Z}_{>0}, \quad Y < Z,$$

be an existing Erdős–Straus witness, written in the complete first-half coordinates of the preceding global-sign theorem. Thus a is the unique smallest denominator,

$$\frac{p}{4} < a < \frac{p}{2}, \quad R = 4a - p, \quad 3 \leq R < p, \quad w = \frac{p}{R}.$$

Every witness lies in exactly one of the following two coordinate channels, up to the retained exchange of the final two denominators:

$$E : (Y, Z) = \left(\frac{pa + a^2/u}{R}, \frac{pa + p^2u}{R} \right), \quad u \mid a^2, \quad R \mid 4u + 1, \quad (E)$$

$$M : (Y, Z) = (pb, pc), \quad b \geq 2, \quad c \geq b + 1, \quad a(4bc - b - c) = pbc. \quad (M)$$

These are the original integral coordinates, not continuously enlarged model domains.

Retain the ordered literal roots

$$\ell = (p, a, Y, Z), \quad S = p + a + Y + Z,$$

and define

$$H(T) = -S^{-1} \prod_{j=0}^3 (T - \ell_j) = AT^4 + T^3 + BT^2 + CT + D, \quad A = -S^{-1}.$$

Put $d_j = H'(\ell_j)$. For each label choose and retain one square root $\xi_j \in \mathbb{C}^\times$ satisfying $\xi_j^2 d_j = 1$. The literal odd receiving map $O : \mathbb{C}_{\text{labels}}^4 \rightarrow \mathbb{C}_q^4$ has columns

$$o_j = \xi_j \begin{pmatrix} 1 \\ -id_j \\ Ad_j^2 + 2\ell_j d_j \\ i(7\ell_j^2 d_j - 13d_j^2) \end{pmatrix}. \quad (1)$$

All norms below are the operator norms induced by the displayed standard Hermitian coordinates. A change of metric is treated separately in Section 5.

Let e_r be the elementary symmetric functions of the four literal roots and set

$$\begin{aligned} \mathfrak{N} = & 60e_3S^5 - 20e_2^2S^4 + 87e_2^3S^2 - 282e_2e_3S^3 - 51e_4S^4 \\ & - 28e_2^4 + 22e_2^2e_3S + 431e_3^2S^2 + 204e_2e_4S^2 \\ & + 224e_2^2e_4 - 168e_2e_3^2 - 856e_3e_4S - 448e_4^2. \end{aligned} \quad (2)$$

The preceding exact reduction of the four rows in (1) gives

$$\det O = -\frac{4e\mathfrak{N}}{S^3}, \quad \epsilon = A^2 \prod_{i < j} (\ell_j - \ell_i) \prod_j \xi_j \in \{1, -1\}. \quad (3)$$

The sign records both root order and all four square-root choices.

Define the exact constants

$$\mathfrak{c}_0 = \frac{81920}{483^6}, \quad \mathfrak{c} = \frac{80}{6279^4} = \frac{81920}{4 \cdot 52^4 \cdot 483^4}. \quad (4)$$

Theorem 1.1 (Sharp prime growth of the literal inverse). *Every positive integral witness in the domain above satisfies*

$$\|O^{-1}\|_2 < \frac{360}{\mathfrak{c}} p\sqrt{w} \leq \frac{120\sqrt{3}}{\mathfrak{c}} p^{3/2}, \quad (5)$$

$$\|\wedge^2 O^{-1}\|_2 < \frac{180}{\mathfrak{c}}, \quad (6)$$

$$\|\wedge^3 O^{-1}\|_2 < \frac{41}{4\mathfrak{c}p^2}, \quad (7)$$

$$|\det O^{-1}| < \frac{1}{4\mathfrak{c}p^2S^3} < \frac{1}{4\mathfrak{c}p^{13/2}}. \quad (8)$$

The exponents in (5)–(7) are optimal over actual prime witnesses.

If $s_1(O) \geq s_2(O) \geq s_3(O) \geq s_4(O) > 0$, then

$$s_3(O) > \sqrt{\mathfrak{c}/180}.$$

Consequently the singular-value decomposition gives

$$O^{-1} = u\varphi + B, \quad \text{rank}(u\varphi) \leq 1, \quad \|B\|_2 < \sqrt{180/\mathfrak{c}}. \quad (9)$$

The line represented by $u\varphi$ may depend on the witness.

2. SHAPE-SENSITIVE DETERMINANT MARGINS

The proof of Theorem 1.1 uses the full coefficient certificates of the global-sign theorem, but extracts information not used by the earlier prime-only estimate.

2.1. Exterior channel. For (E), let $P_E(u, w) \in \mathbb{Z}[u, w]$ be defined by

$$\begin{aligned} -64u^2P_E(u, w) = & \mathfrak{N}(16wu, 4u(w+1), \\ & (w+1)(4wu+w+1), 4wu(w+1+4wu)). \end{aligned} \quad (10)$$

The exact division is part of the supplied certificate. With

$$J_E = 16w^2u^2 + 8w^2u + 28wu + 4u + w^2 + 2w + 1,$$

the literal sum and determinant numerator are

$$\frac{S}{p} = \frac{J_E}{16wu}, \quad -\frac{F}{p^2} = \frac{P_E}{4w^2J_E^6}, \quad F = \frac{\mathfrak{N}}{S^6}. \quad (11)$$

The translated polynomial $P_E(1+X, 1+Y)$ has one positive integer coefficient at every point of the rectangle $0 \leq i \leq 12, 0 \leq j \leq 16$, and its least coefficient is 83886080. The translated polynomial

$$[4w^4J_E(u, w)^6]_{u=1+X, w=1+Y}$$

has nonnegative coefficients in the same rectangle and coefficient sum 64449^6 . The verifier reconstructs both objects from (2) and checks, coefficient by coefficient, that

$$449^6P_E(1+X, 1+Y) - 1310720[4w^4J_E^6]_{u=1+X, w=1+Y} \quad (12)$$

has 221 positive coefficients. Notice that this coefficient statement is in the translated variables. The unshifted P_E has negative coefficients; no unshifted coefficient claim is used.

Since $u, w \geq 1$, (12) gives the pointwise inequality

$$P_E(u, w) > \frac{1310720}{449^6} 4w^4J_E(u, w)^6.$$

The rational coefficient is larger than $16\mathfrak{c}_0$. Using (11) proves

$$-F > \mathfrak{c}_0 p^2 w^2 > \mathfrak{c} p^2 w^2. \quad (13)$$

2.2. Middle channel. Set $b = 2 + X$, $c = 3 + X + Y$, and

$$\begin{aligned} L_M &= 4X^2 + 4XY + 18X + 7Y + 19, \\ J_M &= 8X^3 + 12X^2Y + 4XY^2 + 61X^2 + 57XY + 7Y^2 \\ &\quad + 151X + 63Y + 120. \end{aligned}$$

The exact middle numerator $H_M \in \mathbb{Z}[X, Y]$ is the polynomial defined by

$$-\frac{\mathfrak{N}}{p^8} = \frac{H_M}{L_M^6}, \quad \frac{S}{p} = \frac{J_M}{L_M}, \quad -\frac{F}{p^2} = \frac{H_M}{J_M^6}.$$

Its support consists of all 192 lattice points

$$0 \leq i \leq 18, \quad 0 \leq j \leq 12, \quad i + j \leq 20,$$

and every coefficient is positive, with minimum 81920. Exact expansion also proves that every coefficient of

$$483^6H_M - 81920J_M^6 \quad (14)$$

is positive. Therefore

$$-F > \mathfrak{c}_0 p^2. \quad (15)$$

Put $h = Z - Y = p(c - b)$ and $\tau = h/S$. The product $(b + c)^2H_M = (5 + 2X + Y)^2H_M$ has all 237 coefficients in the enlarged trapezoid

$$0 \leq i \leq 20, \quad 0 \leq j \leq 14, \quad i + j \leq 22,$$

and every coefficient is at least 81920. The comparison polynomial $L_M^4(Y + 1)^2J_M^4$ has nonnegative coefficients, lies in the same trapezoid, and has coefficient sum 452^4483^4 . The verifier checks the stronger exact assertion that every coefficient of

$$452^4483^4(b + c)^2H_M - 81920L_M^4(Y + 1)^2J_M^4 \quad (16)$$

is positive. Since

$$w = \frac{L_M}{b + c}, \quad \tau = \frac{(c - b)L_M}{J_M},$$

this gives

$$-F > \mathfrak{c} p^2 \max\{1, w^2 \tau^2\}. \quad (17)$$

3. COFACTORS AND COMPOUND MATRICES

Put $P = \max(p, Y)$ and $D_* = Z - a$. The reciprocal equation gives

$$Y \leq \frac{2pa}{R} < pw, \quad p \leq P \leq pw, \quad P \leq S. \quad (18)$$

If $Z \leq 2a$, then $Y \leq Z \leq 2a$ would imply $1/a + 1/Y + 1/Z \geq 2/a > 4/p$; hence $Z > 2a$ and $a < D_*$. All four roots are below $Z < 2D_*$, so $S < 8D_*$. At any root one of the three integer differences has magnitude at least $D_*/2$, and the other two have magnitude at least one. Therefore

$$|d_j| > \frac{1}{16}, \quad |\xi_j| < 4. \quad (19)$$

The direct derivative bounds are

$$|d_p|, |d_a| \leq pP, \quad |d_Y| \leq P^2, \quad |d_Z| \leq S^2.$$

Substitution in (1) gives the entry table

	p, a columns	Y column	Z column
0	4	4	4
1	$\sqrt{pP}$	P	S
2	$3p\sqrt{pP}$	$3P^2$	$3S^2$
3	$20pP\sqrt{pP}$	$20P^3$	$20S^3$

(20)

In the exterior channel, each 3×3 cofactor has six determinant terms, each bounded by $60S^3P^{5/2}\sqrt{p}$. Thus

$$\|\text{adj } O\|_2 \leq \|\text{adj } O\|_F \leq 1440S^3P^{5/2}\sqrt{p}.$$

By (3) and (13),

$$\|O^{-1}\|_2 < \frac{360P^{5/2}}{cp^{3/2}w^2} \leq \frac{360}{c}p\sqrt{w}.$$

The middle channel needs its gap and the exact cofactor cancellation. Here

$$w = \frac{4bc}{b+c} - 1, \quad 2b-1 \leq w < 4b, \quad w \geq \frac{19}{5}, \quad (21)$$

and, with $d = c - b \geq 1$,

$$\frac{S}{h} = 1 + \frac{2b+1+a/p}{d} < 2w. \quad (22)$$

The three bounds in (21) retain their exact positive remainders. With $B = b - 2 \geq 0$ and $D = d - 1 \geq 0$,

$$w - (2b-1) = \frac{2bd}{2b+d} > 0, \quad 4b - w = \frac{4b^2 + 2b + d}{2b+d} > 0,$$

and

$$w - \frac{19}{5} = \frac{4(5B^2 + 5BD + 13B + 4D)}{5(2B + D + 5)} \geq 0.$$

After clearing positive denominators, the last strict inequality follows from

$$16b^2d + 16bd^2 - 16bd - 6d^2 - 8b^2 - 6b - 3d > 0 \quad (b \geq 2, d \geq 1).$$

Indeed, with $B = b - 2 \geq 0$ and $D = d - 1 \geq 0$, the left-hand side is exactly

$$16B^2D + 8B^2 + 16BD^2 + 80BD + 26B + 26D^2 + 81D + 11.$$

More precisely, writing $L = 4bc - b - c$ and $\alpha = a/p < 1/2$,

$$w = \frac{L}{b+c}, \quad \frac{S}{h} = 1 + \frac{2b+1+\alpha}{d} < 1 + \frac{2b+3/2}{d},$$

and the difference between $2w$ and the final upper bound is the displayed positive polynomial divided by $2d(2b+d)$. The derivative products give

$$|d_a| > \frac{pY}{8}, \quad |d_p| > \frac{pY}{15}, \quad |d_Y| \geq \frac{Y^2h}{4S}, \quad |d_Z| \geq \frac{Z^2h}{4S}.$$

For completeness, after division by p^2 , the first two left-hand sides are

$$\frac{(1-\alpha)(b-\alpha)(c-\alpha)}{1+\alpha+b+c}, \quad \frac{(1-\alpha)(b-1)(c-1)}{1+\alpha+b+c}.$$

The estimates $b-\alpha > 3b/4$, $b-1 \geq b/2$, $3(c-\alpha) > 1+\alpha+b+c$, and $15(c-1) > 4(1+\alpha+b+c)$ prove the claimed lower bounds. The tail bounds follow from $(b-\alpha)(b-1) > b^2/4$ and $(c-\alpha)(c-1) > c^2/4$. For reference, the two denominator comparisons used here are themselves strict:

$$\begin{aligned} 3(c-\alpha) - (1+\alpha+b+c) &= 2c-b-1-4\alpha > 0, \\ 15(c-1) - 4(1+\alpha+b+c) &= 11c-4b-19-4\alpha > 0. \end{aligned}$$

Together with $b > w/4$, these imply

$$|\xi_j| \leq \frac{12}{p\sqrt{w}}. \quad (23)$$

For three retained columns I , let t be the omitted root and $\Delta_I = \prod_{j < l, j, l \in I} (\ell_l - \ell_j)$. Factoring $\xi_j d_j = \xi_j^{-1}$ from the three columns, and replacing the final polynomial row by

$$(7T^2 - 13H'(T)) + \frac{13}{A}(AH'(T) + 2T) = 7T^2 + \frac{26}{A}T,$$

gives the exact identity

$$\det O_{\{1,2,3\},I} = \left(\prod_{j \in I} \xi_j^{-1} \right) \Delta_I \left[14AB - 64 - 104A \sum_{j \in I} \ell_j - 28A^2 \sum_{\substack{j < l \\ j, l \in I}} \ell_j \ell_l \right]. \quad (24)$$

Using $A = -1/S$, $B = -e_2/S$, and the full set of four roots, the bracket in (24) becomes

$$40 - \frac{76t}{S} - \frac{28t^2}{S^2} - \frac{14e_2}{S^2}, \quad (25)$$

whose modulus is below 151. Moreover,

$$\left| \prod_{j \in I} \xi_j^{-1} \right| = S^{-3/2} |\Delta_I| \sqrt{\prod_{j \in I} |t - \ell_j|}. \quad (26)$$

For an omitted core root, use $|\Delta_I| \leq PSh$ and $\prod_{j \in I} |t - \ell_j| \leq pPS$. For an omitted tail, use $|\Delta_I| \leq pS^2$ and respectively $\prod_{j \in I} |t - \ell_j| \leq P^2h$ or S^2h . Equations (17), (24), and (26) bound every entry in the first inverse column by

$$\frac{108}{\mathfrak{c}} p\sqrt{w}. \quad (27)$$

The constants before 108 in the three omitted-root cases are $151/4$, $151 \cdot 2^{3/2}/4$, and $151/(8\sqrt{2})$.

For the remaining inverse columns, (20), (23), and the baseline part of (17) give

$$\frac{1080w^{3/2}}{cp}, \quad \frac{360\sqrt{w}}{cp^2}, \quad \frac{54}{cp^3\sqrt{w}}.$$

Since $p \geq 13$ and $w \leq p/3$, these are at most 28, 1, 1 times $p\sqrt{w}/\mathfrak{c}$. The Frobenius estimate is therefore

$$\|O^{-1}\|_2 < 2\sqrt{108^2 + 28^2 + 1 + 1} \frac{p\sqrt{w}}{\mathfrak{c}} < \frac{224}{\mathfrak{c}} p\sqrt{w},$$

which is covered by (5).

The entries of $\wedge^2 O^{-1}$ are the complementary 2×2 minors of O , divided by $\det O$, with their retained signs. In the exterior channel, every numerator is at most $120S^3P^2$, hence every inverse minor is below $30/\mathfrak{c}$.

In the middle channel, a tail entry acquires a factor $\sqrt{\tau}$ through

$$\sqrt{|d_Y|} \leq Y\sqrt{\tau}, \quad \sqrt{|d_Z|} \leq Z\sqrt{\tau}.$$

Every core-tail inverse minor is bounded by

$$\frac{30\sqrt{w\tau}}{\mathfrak{c} \max\{1, (w\tau)^2\}} \leq \frac{30}{\mathfrak{c}}, \quad (28)$$

α_j in $s_j(O_p) \asymp p^{\alpha_j}$

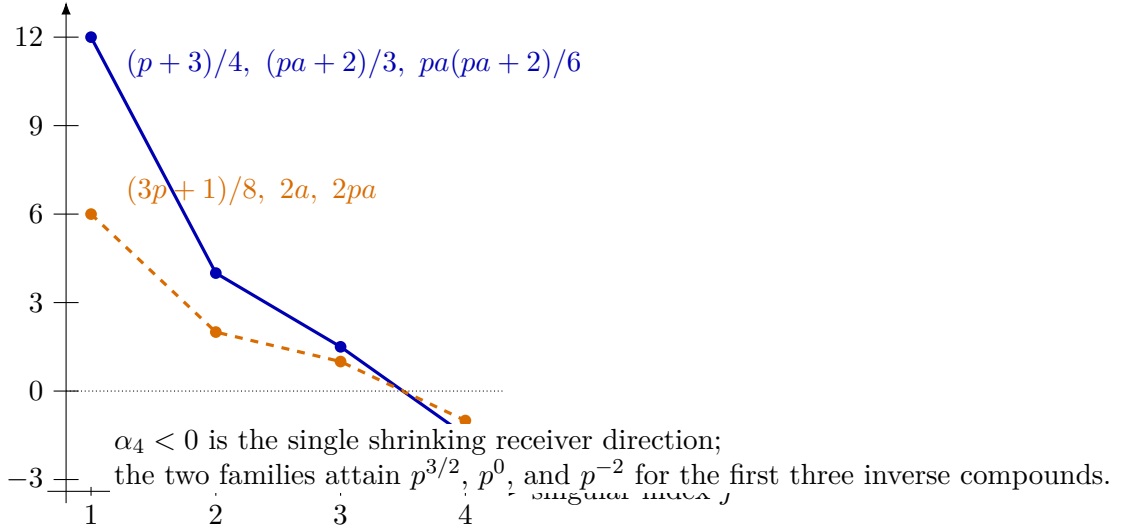


FIGURE 1. Exact singular-value exponents on the two infinite prime families. The first family attains the $p^{3/2}$ inverse exponent and the constant second inverse compound; the second attains the p^{-2} third inverse compound exponent. The figure records powers only; the constants are given in (32) and (38).

and the core-core minors obey the same final bound.

For the tail-tail pair set $Q_c(T) = (T - a)(T - p)$. Then

$$d_Y = \frac{hQ_c(Y)}{S}, \quad d_Z = -\frac{hQ_c(Z)}{S}.$$

The modulus of the rows-(2, 3) minor is exactly

$$\frac{h^2}{S} \sqrt{Q_c(Y)Q_c(Z)} \left| 14YZ + \frac{26}{S} \{YQ_c(Z) + ZQ_c(Y)\} - \frac{7}{S^2} \{Z^2Q_c(Y) + Y^2Q_c(Z)\} \right|. \quad (29)$$

Using $Q_c(Y) < Y^2$, $Q_c(Z) < Z^2$, $Y + Z < S$, and $YZ \leq S^2/4$, the bracket is below $44YZ$; division by (17) gives $11/\mathfrak{c}$. The row pairs (1, 3) and (1, 2) have exact polynomial differences bounded by $20hS$ and $3h$, giving $5/\mathfrak{c}$ and $1/\mathfrak{c}$. Row-0 pairs are below $1/\mathfrak{c}$ by (23). Thus all 36 entries of the six-by-six second inverse compound are below $30/\mathfrak{c}$, and (6) follows from its Frobenius norm.

Finally, (20) gives

$$\|O\|_2^2 \leq \|O\|_F^2 < 64 + 4S^2 + 36S^4 + 1600S^6 < 41^2S^6.$$

For a 4×4 matrix, $\|\wedge^3 O^{-1}\|_2 = \|O\|_2 / |\det O|$, so (7) follows from either determinant margin.

It remains to bound S . In E write $4u + 1 = tR$. Since $R \equiv 3 \pmod{4}$, $t \equiv 3 \pmod{4}$ and $t \geq 3$, while direct substitution gives

$$Z = \frac{p(pt + 1)}{4} > p^{3/2}.$$

In M, $p \nmid a$ and $a(4bc - b - c) = pbc$, so $p \mid 4bc - b - c < 4bc$. Hence $bc > p/4$ and

$$S > p(b + c) \geq 2p\sqrt{bc} > p^{3/2}.$$

This proves (8). Since

$$\|\wedge^2 O^{-1}\|_2 = \frac{1}{s_3(O)s_4(O)} < \frac{180}{\mathfrak{c}}$$

and $s_3 \geq s_4$, one has $s_3 > \sqrt{\mathfrak{c}/180}$. Removing the largest singular summand of O^{-1} leaves a rank-three operator B with norm $1/s_3$, proving (9).

4. TWO INFINITE PRIME FAMILIES AND EXACT LEADING CONSTANTS

4.1. **The family through** (13; 4, 18, 468). For each prime $p \equiv 13 \pmod{24}$, put

$$a = \frac{p+3}{4}, \quad T = pa, \quad Y = \frac{T+2}{3}, \quad Z = \frac{T(T+2)}{6}. \quad (30)$$

Here a is even and $T+2 \equiv 0 \pmod{3}$; hence every displayed quantity is integral. Moreover

$$\frac{1}{Y} + \frac{1}{Z} = \frac{3}{T}, \quad \frac{1}{a} + \frac{3}{pa} = \frac{4}{p}.$$

This is (E) with $R = 3$ and $u = a^2/2$.

Choose positive real square roots for $d_a, d_Y > 0$ and positive-imaginary ones for $d_p, d_Z < 0$; then $\epsilon = 1$. In the root order (p, a, Y, Z) , the exact rational root formulas give the following entrywise asymptotic matrix, with relative error $O(p^{-1})$ in every displayed nonzero entry:

$$O_p = \begin{pmatrix} 4ip^{-3/2} & 4p^{-3/2} & 12p^{-2} & 96ip^{-4} \\ -\frac{1}{4}p^{3/2} & -\frac{i}{4}p^{3/2} & -\frac{i}{12}p^2 & -\frac{1}{96}p^4 \\ -\frac{i}{2}p^{5/2} & \frac{1}{8}p^{5/2} & \frac{1}{72}p^4 & -\frac{i}{3072}p^8 \\ \frac{13}{64}p^{9/2} & -\frac{13i}{64}p^{9/2} & -\frac{i}{288}p^6 & \frac{5}{221184}p^{12} \end{pmatrix}.$$

The complete compound matrices give

j	power of p	leading coefficient of $\ \wedge^j O_p\ _2$
1	12	5/221184
2	16	5/15925248
3	35/2	$5\sqrt{2}/(2^{18}3^5)$
4	16	5/7962624.

(31)

At rank three, two column triples have the same leading order, with coefficients $-5/63700992$ and $-5i/63700992$. Their joint Euclidean norm, not either selected minor, produces the factor $\sqrt{2}$.

Successive compound-norm ratios yield

$$\begin{aligned} s_1(O_p) &= \frac{5}{221184}p^{12}(1 + O(p^{-1})), & s_2(O_p) &= \frac{1}{72}p^4(1 + O(p^{-1})), \\ s_3(O_p) &= \frac{1}{2\sqrt{2}}p^{3/2}(1 + O(p^{-1})), & s_4(O_p) &= 4\sqrt{2}p^{-3/2}(1 + O(p^{-1})). \end{aligned} \quad (32)$$

In particular,

$$p^{-3/2}\|O_p^{-1}\|_2 \longrightarrow \frac{1}{4\sqrt{2}}, \quad \|\wedge^2 O_p^{-1}\|_2 \longrightarrow \frac{1}{2}, \quad p^4\|\wedge^3 O_p^{-1}\|_2 \longrightarrow 36. \quad (33)$$

The leading cofactors give the operator limit

$$p^{-3/2}O_p^{-1} \longrightarrow \frac{1}{8}(-i, 1, 0, 0)^T e_0^*. \quad (34)$$

Thus the smallest right singular vector tends, up to phase, to $v = 2^{-1/2}(-i, 1, 0, 0)^T$, and $p^3 s_4(O_p)^2 \rightarrow 32$. The other three eigenvalues of $p^3 O_p^* O_p$ diverge, so the continuous functional calculus proves, for every fixed $t > 0$,

$$e^{-tp^3 O_p^* O_p} \longrightarrow e^{-32t} v v^*. \quad (35)$$

No statement at $t = 0$ follows from this limit.

4.2. **The family through** (13; 5, 10, 130). For the same prime progression put

$$a = \frac{3p+1}{8}, \quad Y = 2a, \quad Z = 2pa. \quad (36)$$

Then a, Y, Z are positive integers and

$$\frac{1}{a} + \frac{1}{2a} + \frac{1}{2pa} = \frac{4}{p}.$$

This is (E) with $R = (p+1)/2$ and $u = a$, since $4u+1 = 3R$. The derivative signs are $(+, +, -, -)$; choose positive real and positive-imaginary roots accordingly, so $\epsilon = -1$.

The complete leading receiving matrix is

$$\begin{pmatrix} \sqrt{32/5}p^{-1} & 8p^{-1}/\sqrt{15} & i\sqrt{32/3}p^{-1} & 4ip^{-2}/3 \\ -i\sqrt{10}p/8 & -i\sqrt{15}p/8 & -\sqrt{6}p/8 & -3p^2/4 \\ \sqrt{10}p^2/4 & 3\sqrt{15}p^2/32 & -3i\sqrt{6}p^2/16 & -27ip^4/16 \\ 159i\sqrt{10}p^3/256 & -33i\sqrt{15}p^3/128 & 165\sqrt{6}p^3/256 & 135p^6/16 \end{pmatrix},$$

again with relative $O(p^{-1})$ error in each displayed nonzero entry. The full compound norms are

j	power of p	leading coefficient of $\ \wedge^j O_p\ _2$
1	6	135/16
2	8	$135\sqrt{991}/512$
3	9	$675\sqrt{3}/512$
4	8	2565/256.

(37)

Here the relevant small-root row norm, two-row exterior norm, and determinant are exactly $991/1024$, $75/1024$, and $-19/16$, respectively. Therefore

$$\begin{aligned} s_1(O_p) &\sim \frac{135}{16}p^6, & s_2(O_p) &\sim \frac{\sqrt{991}}{32}p^2, \\ s_3(O_p) &\sim \frac{5\sqrt{3}}{\sqrt{991}}p, & s_4(O_p) &\sim \frac{38}{5\sqrt{3}}p^{-1}, \end{aligned} \quad (38)$$

and

$$p^{-1}\|O_p^{-1}\|_2 \rightarrow \frac{5\sqrt{3}}{38}, \quad \|\wedge^2 O_p^{-1}\|_2 \rightarrow \frac{\sqrt{991}}{38}, \quad p^2\|\wedge^3 O_p^{-1}\|_2 \rightarrow \frac{16}{19}. \quad (39)$$

Dirichlet's theorem applies because $\gcd(13, 24) = 1$, so both families contain infinitely many prime witnesses. Equations (33) and (39) prove the optimality of all three exponents in Theorem 1.1.

5. METRICS, CONDUCTOR, AND OBSERVATIONS

Let G and Q be positive-definite Hermitian matrices on $V_\beta = \mathbb{C}_{\text{labels}}^4$ and $E_q = \mathbb{C}_q^4$, respectively. For $1 \leq j \leq 4$, exterior-power submultiplicativity applied to $G^{1/2}O^{-1}Q^{-1/2}$ gives

$$\|\wedge^j O^{-1}\|_{Q \rightarrow G} \leq \left[\frac{\prod_{a=1}^j \lambda_a^\downarrow(G)}{\prod_{a=1}^j \lambda_a^\uparrow(Q)} \right]^{1/2} B_j(p), \quad (40)$$

where B_j is the corresponding bound in (5)–(8). If the metrics depend on p , their eigenvalue factors remain present.

The conductor maps in the source programme have the exact types

$$O : V_\beta \rightarrow E_q, \quad \Phi : E_q \rightarrow U, \quad L_U : U \rightarrow W, \quad \Psi = L_U \Phi : E_q \rightarrow W.$$

On the stated invertible restrictions, the complete composite and inverse are

$$C = L_U \Phi O = \Psi O, \quad C^{-1} = O^{-1} \Phi^{-1} L_U^{-1} = O^{-1} \Psi^{-1}. \quad (41)$$

Thus (9) transports as

$$C^{-1} = B \Psi^{-1} + u(\varphi \Psi^{-1}). \quad (42)$$

The second summand has rank at most one. The first is uniformly bounded only when Ψ^{-1} is fixed or uniformly bounded; the receiver estimate does not remove conductor resonances.

For a further linear observation $\Lambda : E_q \rightarrow Y$, let $Y_0 = \text{im}(\Lambda O) = \text{im } \Lambda$. Then

$$\ker(\Lambda O) = O^{-1} \ker \Lambda, \quad \text{rank}(\Lambda O) = \text{rank } \Lambda, \quad (43)$$

and the quotient covariance on the actual image is

$$Q_{\Lambda O} = \left[(\Lambda O G^{-1} O^* \Lambda^*)|_{Y_0} \right]^{-1}. \quad (44)$$

In an ambient codomain larger than Y_0 , the corresponding formula uses the Moore–Penrose inverse. An observation after the conductor retains every factor:

$$\ker(\Lambda C) = O^{-1} \Phi^{-1} L_U^{-1} \ker \Lambda.$$

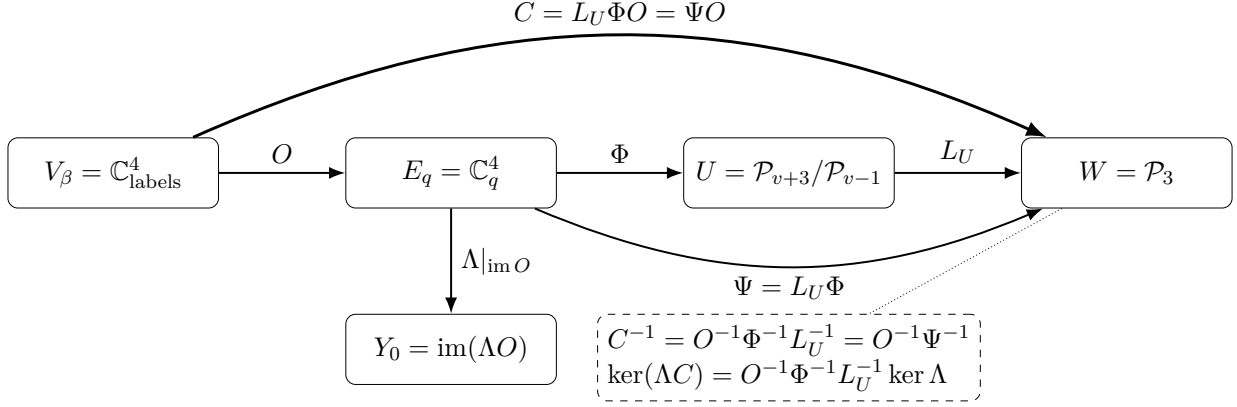


FIGURE 2. The exact typed conductor chain. Omitting Φ makes the composite ill-typed. The receiver's rank-one inverse decomposition passes through Ψ^{-1} , while any growth or resonance of Ψ^{-1} remains.

The metric paired exactly to O is

$$G_{\text{out}} = O^{-*} G_{\text{in}} O^{-1},$$

and through the conductor it is

$$C^{-*} G_{\text{in}} C^{-1} = \Psi^{-*} O^{-*} G_{\text{in}} O^{-1} \Psi^{-1}.$$

Neither Φ nor Ψ is declared unitary. Finally, $\epsilon \det O = 4(-F)S^3 > 0$, but no sign or phase for $\det(\Psi O)$ follows without retaining $\det \Psi$.

6. SCOPE

The theorem begins with an existing integral witness. It proves that the literal receiving stage is invertible there, bounds its inverse and compound maps, and determines sharp growth on two infinite families. It does not produce a witness at a prescribed hard prime and therefore does not prove the Erdős–Straus conjecture. The heat limit (35) belongs to the displayed finite-dimensional ES receiver; it is not an RH heat operator. The conductor conclusions are exactly those of (41)–(42), with all metric and resonance factors retained.

7. ORIGINAL FOUR-LABEL ES RECEIVER: FULL SPECTRUM AND FIRST CORRECTIONS

This section derives the complete singular asymptotics of the literal four-label Erdős–Straus receiver, including its exact inverse, its rank-three exceptional locus, all exterior-power scales, and the relation between inverse rates and total volume. It retains the original denominator labels and every square-root branch. The receiver is the linear extension defined in the cited ES–Fable bridge; it is not asserted to be the Jacobian of the originating nonlinear three-dimensional construction described in Tao's exposition [4].

7.1. ES1. Original roots, chart, signs, and exact inverse. Retain four distinct positive labelled roots $\ell = (p, a, Y, Z)$, the original sum $S = p + a + Y + Z$, and

$$f(x) = \prod_j (x - \ell_j), \quad A = -S^{-1}, \quad H = Af, \quad d_j = Af'(\ell_j), \quad \xi_j^2 d_j = 1. \quad (\text{ES1})$$

The source and target coefficient frames are exactly the four labels and the four displayed receiving coordinates. Their Euclidean receiver has columns

$$o_j = \xi_j (1, -id_j, Ad_j^2 + 2\ell_j d_j, i(7\ell_j^2 d_j - 13d_j^2))^T. \quad (\text{ES2})$$

Thus every independent square-root sign is retained. Changing a sign is right multiplication by its diagonal sign isometry and changes the corresponding label coordinate of every inverse and observed vector.

Let e_2, e_3 be the elementary symmetric sums of the literal roots, and put

$$\begin{aligned} P_2(x) &= x^2 + \frac{19S}{7}x + \frac{e_2}{2} - \frac{10S^2}{7}, & \mathfrak{h} &= \sum_j \frac{P_2(\ell_j)}{d_j^2}, \\ R_1(x) &= i \left[x^3 + \left(\frac{19S^2}{7} + e_2 \right) x - \frac{10S^3}{7} + \frac{Se_2}{2} - \frac{3e_3}{4} \right], \\ R_2(x) &= \frac{5S^2 - 13Sx}{7}, & R_3(x) &= \frac{i(S - 4x)}{28}, \\ \mathfrak{r}_\nu &= \sum_j \frac{R_\nu(\ell_j)}{d_j^2}. \end{aligned} \quad (\text{ES3})$$

On the exact invertibility locus calculated in ES5, the supplied rational expressions are the actual inverse of ES2:

$$(O^{-1}e_0)_j = \xi_j^{-1} \frac{P_2(\ell_j)/d_j^2}{\mathfrak{h}}, \quad (O^{-1})_{j\nu} = \frac{A\xi_j^{-1}}{d_j^2} \left[R_\nu(\ell_j) - P_2(\ell_j) \frac{\mathfrak{r}_\nu}{\mathfrak{h}} \right], \quad 1 \leq \nu \leq 3. \quad (\text{ES4})$$

Here is a direct derivation, retaining all denominators. For target b , put $z_j = \xi_j(O^{-1}b)_j$ and $t_j = d_j z_j$. There is a unique polynomial $L(x) = l_3 x^3 + l_2 x^2 + l_1 x + l_0$ with $t_j = L(\ell_j)/f'(\ell_j)$. The coefficients of the Lagrange identity, or polynomial division by the monic f , give

$$\sum_j \ell_j^n / f'(\ell_j) = 0, 0, 0, 1, S, S^2 - e_2 \quad (0 \leq n \leq 5).$$

Writing $U_L = \sum_j L(\ell_j)$, the last three receiver equations become exactly

$$\begin{aligned} l_3 &= ib_1, \quad A^2 U_L + 2(l_2 + Sl_3) = b_2, \quad 7[l_1 + Sl_2 + (S^2 - e_2)l_3] - 13AU_L = -ib_3, \\ U_L &= l_3(S^3 - 3Se_2 + 3e_3) + l_2(S^2 - 2e_2) + Sl_1 + 4l_0. \end{aligned}$$

The first equation fixes l_3 ; the coefficient determinant in (l_1, l_0) of the other two equations is $(1/S)(52/S) - (4/S^2)20 = -28/S^2 \neq 0$. Hence their homogeneous solution space has dimension one. Substitution verifies that their complete solution is $L = cP_2 + \sum_{\nu=1}^3 b_\nu R_\nu$. The remaining first equation is $A \sum_j L(\ell_j)/d_j^2 = b_0$, which determines $c = (b_0/A - \sum b_\nu \mathfrak{r}_\nu)/\mathfrak{h}$ and proves ES4. This is an exact connecting map from the supplied auxiliary polynomial column to the original labelled inverse.

For the determinant put $\Delta_\ell = \prod_{i < j} (\ell_j - \ell_i)$. Factoring each column by d_j , applying the same polynomial coefficient elimination, and using the Vandermonde determinant gives

$$\det O = 28\epsilon A \left(\prod_j d_j \right) \mathfrak{h}, \quad \epsilon = A^2 \Delta_\ell \prod_j \xi_j \in \{1, -1\}. \quad (\text{ES5})$$

Indeed $\prod_j f'(\ell_j) = \Delta_\ell^2$ for four roots, so $\epsilon^2 = 1$. Equivalently, before any square-root choices, the determinant of the four unscaled columns in ES2 is $28A^3 \Delta_\ell (\prod d_j) \mathfrak{h}$. Thus among all distinct positive labelled roots the exact exceptional receiver space is $\mathfrak{h} = 0$; its complement is precisely the domain of ES4. Below, nonvanishing is proved directly throughout the large- w regime used by the asymptotics. No unproved global certificate at every smaller integral witness is needed for that calculation.

To verify the determinant coefficient explicitly, interpolate $R_0(\ell_j) = 1/d_j$, writing $R_0 = r_0 + r_1 x + r_2 x^2 + r_3 x^3$. After factoring the d_j and cancelling the row constants $(-i)i = 1$, the other three row polynomials are 1, $A^2 f' + 2x$, and $7x^2 - 13A f'$. The determinant of their four coefficient rows is

$$\frac{28}{S^2} r_1 + \frac{104}{S} r_2 + \left(64 - \frac{14e_2}{S^2} \right) r_3 = 28A^2 \sum_j \frac{P_2(\ell_j) R_0(\ell_j)}{f'(\ell_j)} = 28A^3 \mathfrak{h}.$$

The equality uses exactly the Lagrange sums above. Multiplication by the evaluation determinant Δ_ℓ , by $\prod d_j$, and finally by $\prod \xi_j$ proves ES5 with its stated sign.

The exceptional space also has exact connecting data. At $\mathfrak{h} = 0$, the last three receiver rows still have rank three by the coefficient determinant above. Hence O has rank exactly three, with kernel generated by the nonzero vector $(\xi_j^{-1}P_2(\ell_j)/d_j^2)_j$. Its image is precisely the hyperplane

$$\{b \in \mathbb{C}^4 : b_0/A - \mathfrak{r}_1 b_1 - \mathfrak{r}_2 b_2 - \mathfrak{r}_3 b_3 = 0\}.$$

The defining cokernel row is nonzero because $A \neq 0$. This follows from the first-row compatibility equation after solving all last-three-row equations. Thus the exact defect determines both kernel and cokernel and is not merely a failed inverse formula.

7.2. ES2. Uniform large- w exterior spectrum. In the original exterior chart retain

$$\begin{aligned} R = 4a - p > 0, \quad w = p/R, \quad t = a/p = (1 + w^{-1})/4, \quad u \geq 1, \\ Y = pw(t + t^2/u), \quad Z = pw(t + u), \quad \frac{Y}{Z} = \frac{t}{u}, \quad \frac{YZ}{Y + Z} = pwt. \end{aligned} \quad (\text{ES6})$$

These equalities are literal algebraic identities; integrality remains an additional property of the original witness, not of an arbitrary real chart point. For $w \geq 256$, they imply $a < p < Y < Z$, $Y \geq pw/4$, $Z \geq pw$. Set $U = Y + Z$, $\tau = (Z - Y)/U$. Since $Y/Z = t/u \leq 257/1024$, one has $1/2 < \tau < 1$.

Choose $\xi_p = i/\sqrt{-d_p}$, $\xi_a = 1/\sqrt{d_a}$, $\xi_Y = 1/\sqrt{d_Y}$, $\xi_Z = i/\sqrt{-d_Z}$. The exact derivative equations include

$$\begin{aligned} -d_p = p^2 wt(1 - t) \frac{(1 - p/Y)(1 - p/Z)}{1 + (p + a)/U}, \quad d_a = p^2 wt(1 - t) \frac{(1 - a/Y)(1 - a/Z)}{1 + (p + a)/U}, \\ d_Y = Y^2 \tau \frac{(1 - p/Y)(1 - a/Y)}{1 + (p + a)/U}, \quad -d_Z = Z^2 \tau \frac{(1 - p/Z)(1 - a/Z)}{1 + (p + a)/U}. \end{aligned} \quad (\text{ES7})$$

The bounds $p/Y \leq 4/w$, $p/Z \leq 1/w$, $(p + a)/U \leq 4/(3w)$ prove, with constants independent of $u \geq 1$,

$$-d_p, d_a = \frac{3}{16} p^2 w [1 + O(w^{-1})], \quad d_Y = Y^2 \tau [1 + O(w^{-1})], \quad -d_Z = Z^2 \tau [1 + O(w^{-1})]. \quad (\text{ES8})$$

All estimates in this section are uniform for $p \geq 1, u \geq 1, w \geq 256$; no nearly coincident-tail denominator is hidden in them.

For completeness $\mathfrak{h} < 0$ on this entire regime. Positivity of the four roots gives $e_2 \leq 3S^2/8$, while $p/S \leq 1/w$. Consequently both core values satisfy $P_2(p), P_2(a) \leq -S^2$. ES7 gives $|d_p|, d_a \leq p^2 w/4$, so the core contribution to $\mathfrak{h}$ is at most $-32S^2/(p^4 w^2)$. At every root $|P_2(\ell_j)| \leq 6S^2$. ES7 gives $d_Y \geq Y^2/3$, $-d_Z \geq Z^2/3$, hence the absolute tail contribution is at most

$$54S^2(Y^{-4} + Z^{-4}) \leq 27648S^2/(p^4 w^4).$$

The ratio of this last bound to the core magnitude is at most $864/w^2 < 1$. This proves nonvanishing of ES4–5 directly, without relying on an unexecuted global coefficient certificate.

Define the positive literal-tail quantities

$$\begin{aligned} A_3 = \sqrt{\tau[Y^6(7 - 13\tau)^2 + Z^6(7 + 13\tau)^2]}, \\ A_{23} = \tau^2 Y^2 Z^2 U \left(40 - \frac{14YZ}{U^2} \right). \end{aligned} \quad (\text{ES9})$$

The bracket is at least $73/2$. The complete increasingly numbered by decreasing size singular values are

$$\begin{aligned} s_1 = A_3[1 + O(w^{-1})], \quad s_2 = (A_{23}/A_3)[1 + O(w^{-1})], \\ s_3 = \sqrt{3/8} p \sqrt{w} [1 + O(w^{-1})], \quad s_4 = \sqrt{32/3} (p \sqrt{w})^{-1} [1 + O(w^{-1})]. \end{aligned} \quad (\text{ES10})$$

Here and below singular values use the original Euclidean label and receiving metrics, until the explicit transport ES33–37 in `ES_FAMILY_CORRECTIONS.md`.

We prove all four rates by the complete exterior maps. The leading tail block in rows 2 and 3 is

$$\begin{pmatrix} Y^2 \sqrt{\tau}(2 - Y\tau/U) & -iZ^2 \sqrt{\tau}(2 + Z\tau/U) \\ iY^3 \sqrt{\tau}(7 - 13\tau) & Z^3 \sqrt{\tau}(7 + 13\tau) \end{pmatrix}. \quad (\text{ES11})$$

Its determinant is exactly A_{23} : expanding the two terms reduces the remaining bracket to $\tau U(40 - 14YZ/U^2)$. ES8 gives errors bounded entrywise by $CY^2/w, CZ^2/w, CY^3/w, CZ^3/w$, respectively. The last-row norm is $A_3[1 + O(w^{-1})]$; its Z term is uniformly nonzero even if $7 - 13\tau$ vanishes.

Here are the bounds for every omitted exterior coefficient. Core entries in rows 0,1,2,3 have respective sizes at most $C/(p\sqrt{w}), Cp\sqrt{w}, Cp^2\sqrt{w}, Cp^3w^{3/2}$; tail entries in those rows have respective sizes at most $C/L, CL, CL^2, CL^3$, for $L = Y, Z$. Relative to $A_{23} \geq (73/8)Y^2Z^3$, any other two-column minor is bounded by C/w : for tail columns lowering row 2 costs at most C/Y , and for one core column the largest ratio is bounded by $C[p^2\sqrt{w}/Y^2 + p^3w^{3/2}/(Y^2Z)]$. Both are at most C/w ; two core columns are smaller. The finite number of minors then bounds the entire second exterior map, not only one chosen coefficient.

At exterior rank three the two leading entries, in rows 1,2,3 and columns (p, Y, Z) , (a, Y, Z) , are

$$-\frac{\sqrt{3}}{4}p\sqrt{w}A_{23}[1 + O(w^{-1})], \quad -\frac{i\sqrt{3}}{4}p\sqrt{w}A_{23}[1 + O(w^{-1})].$$

The other terms in either same minor cost at most $Cp/Y + Cp^2w/(YZ) = O(w^{-1})$ relatively. A triple with two core columns has relative size at most $Cw^{-3/2}$, and replacing row 1 by row 0 costs at most $C/(p^2w)$. Thus the leading exterior map has one nonzero row with both displayed entries, giving

$$\|\wedge^3 O\| = \sqrt{3/8}p\sqrt{w}A_{23}[1 + O(w^{-1})]. \quad (\text{ES12})$$

For the full determinant the core rows 0,1 minor equals $(d_a - d_p)/\sqrt{-d_p d_a} = 2 + O(w^{-1})$. The largest alternative pairing, core rows 0,2 and tail rows 1,3, has relative size at most Cp/Y ; core rows 0,3 and tail rows 1,2 cost at most $Cp^2w/(YZ)$. The remaining pairings cost at most these bounds. Therefore

$$|\det O| = 2A_{23}[1 + O(w^{-1})]. \quad (\text{ES13})$$

The identities $\|\wedge^j O\| = s_1 \cdots s_j$ now prove ES10 with all four singular values retained.

The same calculation, or ES4 with ES7, gives the inverse operator limit

$$\frac{O^{-1}}{p\sqrt{w}} = \frac{\sqrt{3}}{8}(-i, 1, 0, 0)^\top e_0^* + O(w^{-1}) \quad (\text{ES14})$$

in operator norm, uniformly in the stated regime. To see why all columns are controlled, the cofactor of column zero has the two ES12 leading triples, while every cofactor of column one has norm at most $CA_{23}/(p\sqrt{w})$; columns two and three have at most $CA_{23}/(p^2\sqrt{w})$ and $CA_{23}/(p^3w^{3/2})$, respectively. Division by ES13 and then by $p\sqrt{w}$ makes all these columns $O(w^{-1})$, uniformly for $p \geq 1$. The leading two zero-column cofactors have the phases in ES14; their exact formula is ES4.

Let $v_0 = 2^{-1/2}(-i, 1, 0, 0)^\top$. Rank-one singular perturbation of ES14 and ES10 prove, for every fixed $\tau_0 > 0$,

$$e^{-\tau_0 p^2 w O^* O} \longrightarrow e^{-(32/3)\tau_0} v_0 v_0^*, \quad (\text{ES15})$$

uniformly as $w \rightarrow \infty$ in this original chart. The other three exponents diverge because $p^2 w s_3^2 \asymp p^4 w^2 \rightarrow \infty$. The four marked labels remain distinct in the limiting projection.

8. ORIGINAL ES FAMILY: COMPLETE FIRST CORRECTIONS AND OBSERVATION HIERARCHY

This section continues equations ES1–ES15. Every root, map, branch, and denominator is the one fixed there; no relabelling or metric replacement is made.

8.1. ES3. Exact fixed-c integral family and its progression. Fix an integer $c \geq 2$, $c \equiv 2 \pmod{3}$. At primes satisfying $p \equiv 1 \pmod{12}$, $c \mid a = (p+3)/4$, retain

$$T = pa, \quad Y = (T+c)/3, \quad Z = T(T+c)/(3c). \quad (\text{ES16})$$

Since $p \equiv a \equiv 1 \pmod{3}$, $T+c \equiv 0 \pmod{3}$; since $c \mid T$, both Y, Z are integers. Direct fractions give $Y^{-1} + Z^{-1} = 3/T$ and $a^{-1} + 3/(pa) = 4/p$. Thus this is an existing integral ES witness in the original chart, with $R = 3, w = p/3, u = a^2/c \geq 1$ eventually. No occupancy assertion for every prime in a residue class is made.

The explicit choice $c = 29$, $p \equiv 3361 \pmod{24360}$ satisfies $p \equiv 1 \pmod{840}$, $p \equiv -3 \pmod{116}$, and $\gcd(3361, 24360) = 1$. The classical coprime-progression theorem of P. G. Lejeune-Dirichlet gives infinitely many primes in this progression. The retained source is the 1837 paper in Ralf Stephan's English translation, arXiv:0808.1408v2; its opening theorem and translator credit were read directly in the retained TeX. This translation is not described as Dirichlet's original German typesetting. The initial prime 3361 gives exactly

$$(p; a, Y, Z) = (3361; 841, 942210, 91836266490), \quad T = 2826601. \quad (\text{ES17})$$

8.2. ES4. Full analytic matrix and every first singular correction. Use $z = 1/p$ here, to distinguish the expansion variable from the chart's $t = a/p$. Define the exact weighted matrix

$$B(z) = \text{diag}(1, p^{-3}, p^{-4}, p^{-6}) O_p \text{diag}(p^{3/2}, p^{3/2}, 1, p^{-6}). \quad (\text{ES18})$$

The roots and d_j are rational functions of z . Their four derivative leading terms are

$$-\frac{p^3}{16}, \quad \frac{p^3}{16}, \quad \frac{p^4}{144}, \quad -\frac{p^8}{2304c^2}.$$

Factoring these terms out of the prescribed square roots leaves analytic functions equal to one at zero. Every exponent in ES18 is then integral and nonnegative. Hence B is analytic at zero, with a convergent local Taylor series for each fixed c .

Direct substitution gives its complete first two coefficients:

$$B_0 = \begin{pmatrix} 4i & 4 & 0 & 0 \\ -1/4 & -i/4 & 0 & 0 \\ -i/2 & 1/8 & 1/72 & 0 \\ 13/64 & -13i/64 & -i/288 & 5/(27648c^3) \end{pmatrix},$$

$$B_1 = \begin{pmatrix} 20i & 2 & 0 & 0 \\ 5/4 & i/8 & -i/12 & 0 \\ 5i/2 & 5/16 & -1/48 & 0 \\ -83/64 & 53i/128 & 31i/288 & 5/(1536c^3) \end{pmatrix}, \quad B(z) = B_0 + zB_1 + O_c(z^2). \quad (\text{ES19})$$

In particular $\det B_0 = 5/(995328c^3) > 0$. Its inverse is analytic, and

$$B(z)^{-1}e_0 = z_0 + zz_1 + O_c(z^2),$$

$$z_0 = (-i/8, 1/8, 27/8, 1728ic^3/5)^\top, \quad z_1 = (-i/2, -19/16, 135/16, -50112ic^3/5)^\top. \quad (\text{ES20})$$

These are verified by $B_0 z_0 = e_0$, $B_0 z_1 = -B_1 z_0$, giving their exact derivation without an inferred inverse rate. In particular the nonzero entry $(B_1)_{1Y} = -i/12$ and the original Y-coordinate $27/8$ both enter the two core derivatives.

The exact inverse factorization is

$$O_p^{-1} = \text{diag}(p^{3/2}, p^{3/2}, 1, p^{-6}) B(1/p)^{-1} \text{diag}(1, p^{-3}, p^{-4}, p^{-6}). \quad (\text{ES21})$$

Thus its first column has all four expansions

$$(O_p^{-1}e_0)_p = -\frac{i}{8}p^{3/2} - \frac{i}{2}p^{1/2} + O_c(p^{-1/2}), \quad (O_p^{-1}e_0)_a = \frac{1}{8}p^{3/2} - \frac{19}{16}p^{1/2} + O_c(p^{-1/2}),$$

$$(O_p^{-1}e_0)_Y = \frac{27}{8} \left(1 + \frac{5}{2p} + O_c(p^{-2})\right), \quad (O_p^{-1}e_0)_Z = \frac{1728ic^3}{5}p^{-6} \left(1 - \frac{29}{p} + O_c(p^{-2})\right). \quad (\text{ES22})$$

These expand the exact ES4 column, with the same roots, derivative denominators and square-root choices. They are not a different auxiliary receiver.

Every singular value has its first correction evaluated:

$$\boxed{\begin{aligned} s_1(O_p) &= \frac{5}{27648c^3} p^{12} \left(1 + \frac{18}{p} + O_c(p^{-2}) \right), \\ s_2(O_p) &= \frac{1}{72} p^4 \left(1 - \frac{3}{2p} + O_c(p^{-2}) \right), \\ s_3(O_p) &= \frac{1}{2\sqrt{2}} p^{3/2} \left(1 - \frac{41}{4p} + O_c(p^{-2}) \right), \\ s_4(O_p) &= 4\sqrt{2} p^{-3/2} \left(1 + \frac{11}{4p} + O_c(p^{-2}) \right). \end{aligned}} \quad (\text{ES23})$$

The complete determinant, with the retained branches, is

$$\boxed{\det O_p = \frac{5}{995328c^3} p^{16} \left(1 + \frac{9}{p} + O_c(p^{-2}) \right).} \quad (\text{ES24})$$

We prove the errors as well as the coefficients. The leading entry of O_p is row 3, column Z, namely $p^{12}B_{3Z}$; every other entry is at most $O_c(p^8)$. Thus its norm has the relative first correction 18 by ES19. For exterior rank two, the leading minor is rows 2,3 and columns Y,Z:

$$p^{16} \det B_{\{2,3\},\{Y,Z\}} = \frac{5}{1990656c^3} p^{16} \left(1 + \frac{33}{2p} + O_c(p^{-2}) \right).$$

Every other minor is at most $O_c(p^{15})$. Its squared Frobenius norm is therefore the squared leading minor times $1 + O_c(p^{-2})$; the operator norm lies between that leading modulus and the Frobenius norm. This proves the stated first correction for s_1s_2 .

For exterior rank three, both leading minors are in rows 1,2,3 and columns (p, Y, Z) , (a, Y, Z) , of scale $p^{35/2}$. Their exact analytic coefficients are the corresponding minors $M_p(z), M_a(z)$ of B . Expanding ES19 gives

$$\frac{\frac{d}{dz}(|M_p(z)|^2 + |M_a(z)|^2)}{2(|M_p(0)|^2 + |M_a(0)|^2)} \Big|_{z=0} = \frac{25}{4}.$$

The leading norm is $5p^{35/2}/(3981312\sqrt{2}c^3)$. Every other entry is at most $O_c(p^{16})$, so its squared contribution is relatively $O_c(p^{-3})$. The norm of the complete leading row is a lower bound, and the Frobenius norm supplies the matching upper bound. Thus the first correction to $s_1s_2s_3$ is $25/4$. Finally $\det O_p = p^{16} \det B(1/p)$, and $\text{Tr}(B_0^{-1}B_1) = 9$. Taking successive exterior ratios gives $18, 33/2 - 18 = -3/2, 25/4 - 33/2 = -41/4, 9 - 25/4 = 11/4$, proving ES23–24.

A second route to the smallest singular value uses

$$u_0 = \frac{(-i, 1, 0, 0)^\top}{8}, \quad u_1 = (-i/2, -19/16, 0, 0)^\top.$$

ES21 gives the first normalized inverse column $u_0 + p^{-1}u_1 + O_c(p^{-3/2})$; its $p^{-3/2}$ term is in the orthogonal Y coordinate. The other normalized inverse columns are $O_c(p^{-3})$. Since $\|u_0\|^2 = 1/32$ and $2\Re(u_0^*u_1) = -11/64$,

$$p^{-3}\|O_p^{-1}\|^2 = \frac{1}{32} - \frac{11}{64p} + O_c(p^{-2}), \quad \boxed{p^3s_4(O_p)^2 = 32 + \frac{176}{p} + O_c(p^{-2})}. \quad (\text{ES25})$$

8.3. ES5. Exact slow-vector hierarchy and scalar observations. Let v_p be the right singular vector of O_p for s_4 , with its a coordinate positive. Put $v_0 = (-i, 1, 0, 0)^\top/\sqrt{2}$, $w_0 = (i, 1, 0, 0)^\top/\sqrt{2}$. Then

$$v_p = v_0 - \frac{27}{4p}w_0 + \frac{27\sqrt{2}}{2p^{3/2}}e_Y + O_c(p^{-2}). \quad (\text{ES26})$$

Both tail coordinates have stronger relative expansions:

$$\boxed{e_Y^*v_p = \frac{27\sqrt{2}}{2}p^{-3/2} \left(1 + \frac{21}{4p} + O_c(p^{-2}) \right),}$$

$$e_Z^* v_p = \frac{6912\sqrt{2}ic^3}{5} p^{-15/2} \left(1 - \frac{105}{4p} + O_c(p^{-2}) \right). \quad (\text{ES27})$$

To prove relative accuracy, write exactly

$$p^{-3/2} O_p^{-1} = D(z) B(z)^{-1} E(z), \quad D(z) = \text{diag}(1, 1, z^{3/2}, z^{15/2}), \quad E(z) = \text{diag}(1, z^3, z^4, z^6).$$

The right Gram of this inverse has a simple leading eigenvalue $1/32$. Its top eigenvector has coordinates $(1, O_c(z^3), O_c(z^4), O_c(z^6))$, up to scalar normalization: solve the three lower block equations, whose scalar eigenvalue stays separated from zero. Multiplication by E changes the coefficient vector from e_0 by $O_c(z^6)$. The left singular vector before normalization is therefore $D(z)[B(z)^{-1}e_0 + O_c(z^6)]$, preserving the powers in each tail coordinate. Its norm is $\|u_0\|[1 - 11z/4 + O_c(z^2)]$. ES20–22 give ES26 and the relative coefficients $5/2 + 11/4 = 21/4$, $-29 + 11/4 = -105/4$ in ES27. A tiny component is not inferred from a larger vector error.

For $\mathcal{H}_p(\tau) = e^{-\tau p^3 O_p^* O_p}$, the three faster directions are $O_c(e^{-c_1 \tau p^6})$, while ES25 supplies the slow exponential. For each fixed nonzero scalar row ℓ , the following tests are exhaustive:

Exact original-label test	Power ν	Limit of $p^\nu \ell \mathcal{H}_p(\tau) \ell^*$
$\ell v_0 \neq 0$	0	$ \ell v_0 ^2 e^{-32\tau}$
$\ell v_0 = 0, \ell w_0 \neq 0$	2	$(729/16) \ell w_0 ^2 e^{-32\tau}$
$\ell v_0 = \ell w_0 = 0, \ell e_Y \neq 0$	3	$(729/2) \ell e_Y ^2 e^{-32\tau}$
$\ell v_0 = \ell w_0 = \ell e_Y = 0$	15	$(95551488c^6/25) \ell e_Z ^2 e^{-32\tau}$

The vectors v_0, w_0 span exactly the original core coordinates, so no nonzero row is missing. Convergence is locally uniform in $\tau \in (0, \infty)$, without an assertion at zero time. The two pure tail observations also have evaluated first corrections:

$$\begin{aligned} p^3 e_Y^* \mathcal{H}_p(\tau) e_Y &= \frac{729}{2} e^{-32\tau} \left[1 + \frac{21/2 - 176\tau}{p} + O_c(p^{-2}) \right], \\ p^{15} e_Z^* \mathcal{H}_p(\tau) e_Z &= \frac{95551488c^6}{25} e^{-32\tau} \left[1 - \frac{105/2 + 176\tau}{p} + O_c(p^{-2}) \right]. \end{aligned} \quad (\text{ES28})$$

These follow by squaring ES27 and multiplying the exponential from ES25; the faster modes remain below the remainder after either polynomial rescaling.

For the actual two-row observation $L_{\text{obs}} = \begin{pmatrix} 1/\sqrt{2} & i/\sqrt{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$, $D_p = \text{diag}(p, p^{3/2})$, its first row annihilates v_0 and takes value i on w_0 . Hence

$$D_p L_{\text{obs}} \mathcal{H}_p(\tau) L_{\text{obs}}^* D_p \longrightarrow e^{-32\tau} \begin{pmatrix} 729/16 & -729i\sqrt{2}/8 \\ 729i\sqrt{2}/8 & 729/2 \end{pmatrix}. \quad (\text{ES29})$$

Replacing the core phases by their moduli before applying this observation would destroy its cancellation.

8.4. ES6. Same inverse limits, different complete volume on the same map. For primes $p \equiv 13 \pmod{24}$, retain the exact original-chart family

$$a = (p+3)/4, \quad Y = (p+3)(3p+1)/32, \quad Z = p(3p+1)/4. \quad (\text{ES30})$$

Writing $p = 24j + 13$ gives $p+3 = 8(3j+2)$, $3p+1 = 8(9j+5)$, proving integrality. Direct substitution gives $Y^{-1} + Z^{-1} = 12/[p(p+3)] = 3/(pa)$. Thus the same ES identity and receiver ES2 apply, with $R = 3, w = p/3, u = 2$; the initial witness is $(13; 4, 20, 130)$.

Here $Y/p^2 \rightarrow 3/32$, $Z/p^2 \rightarrow 3/4$, $\tau \rightarrow 7/9$. Substituting into the literal ES9 quantities gives

$$\begin{aligned} s_1 &= \frac{7\sqrt{55508999}}{8192} p^6 [1 + O(p^{-1})], \quad s_2 = \frac{2737}{24\sqrt{55508999}} p^4 [1 + O(p^{-1})], \\ s_3 &= \frac{1}{2\sqrt{2}} p^{3/2} [1 + O(p^{-1})], \quad s_4 = 4\sqrt{2} p^{-3/2} [1 + O(p^{-1})], \quad |\det O_p| = \frac{19159}{98304} p^{10} [1 + O(p^{-1})]. \end{aligned} \quad (\text{ES31})$$

Both families have the same limits

$$p^{-3/2}\|O_p^{-1}\| \rightarrow 1/(4\sqrt{2}), \quad \|\wedge^2 O_p^{-1}\| \rightarrow 1/2, \quad e^{-\tau p^3 O_p^* O_p} \rightarrow e^{-32\tau} v_0 v_0^*. \quad (\text{ES32})$$

Their complete determinants have different powers, p^{10} and p^{16} . This compares existing integral families under the same receiver; it does not invent an unrelated spectrum. Their exact relation is $|\det O| = (s_1 s_2) / \|\wedge^2 O^{-1}\|$: the two inverse limits control s_3, s_4 , while the original tail factor $A_{23} = s_1 s_2 [1 + O(w^{-1})]$ supplies the missing volume.

8.5. ES7. Exact fixed-metric and conductor transport. For fixed positive label metric G and fixed positive receiving metric Q , the original energy is $H_{G,Q} = G^{-1} O_p^* Q O_p$. Its orthonormal representative is $\hat{O} = Q^{1/2} O_p G^{-1/2}$, with exact inverse $G^{1/2} O_p^{-1} Q^{-1/2}$. Put $n_G = u_0^* G u_0 > 0$, $qQ = e_0^* Q^{-1} e_0 > 0$. ES21 gives the rank-one inverse limit $G^{1/2} u_0 e_0^* Q^{-1/2}$, whence

$$p^3 \lambda_{\min}(H_{G,Q}) \rightarrow (n_G qQ)^{-1}. \quad (\text{ES33})$$

Its metric-dependent first correction is

$$p^3 \lambda_{\min}(H_{G,Q}) = \frac{1}{n_G qQ} \left[1 - \frac{2\Re(u_0^* G u_1)}{n_G p} + O_{c,G,Q}(p^{-3/2}) \right]. \quad (\text{ES34})$$

Indeed the first inverse-column squared G norm is $n_G + 2\Re(u_0^* G u_1)/p + O(p^{-3/2})$. Other inverse columns enter the weighted target contraction first at order p^{-3} . The simple leading eigenvalue expansion, obtained from its complementary block Schur equation, yields ES34. A $p^{-3/2}$ term may occur when G has nonzero core-Y entries; the stronger Euclidean remainder is not assigned to arbitrary metrics.

The observed covariance $\ell e^{-\tau p^3 H_{G,Q}} G^{-1} \ell^*$ has the same four powers 0, 2, 3, 15, tested successively by $\ell u_0, \ell u_1, \ell e_Y, \ell e_Z$. Its respective leading numerators are

$$|\ell u_0|^2, \quad |\ell u_1|^2, \quad |\ell(27e_Y/8)|^2, \quad |\ell(1728ic^3 e_Z/5)|^2, \quad (\text{ES35})$$

each divided by n_G and multiplied by $e^{-\tau/(n_G qQ)}$, after the preceding numerators vanish. For relative last-coordinate accuracy, write the inverse exactly as $G^{1/2} D(z) B(z)^{-1} E(z) Q^{-1/2}$. Its top left singular vector, returned to label coordinates, is $D(z)[B(z)^{-1} e_0 + O_{G,Q}(z^3)]$, up to normalization: the off-diagonal terms in $E Q^{-1} E$ first occur at order z^3 . The Y and Z coordinate powers therefore stay $z^{3/2}, z^{15/2}$, with the same nonzero leading coefficients. Normalization terms multiply earlier vectors and cancel under the successive annihilation tests. This proves the weighted hierarchy without imposing Euclidean orthogonality after transport.

The attained scalar observation metric is $Q_\ell = (\ell G^{-1} \ell^*)^{-1}$; multiply the covariance by it to obtain measured scalar heat. On a fixed admitted invertible conductor restriction L , take the actual metric $Q = L^* Q_\Gamma L$. Then $qQ = e_0^* (L^* Q_\Gamma L)^{-1} e_0$ retains the full conductor inverse, Gamma mass, and period data.

For moving metrics or conductor restrictions, the underlying error transport is

$$\begin{aligned} & \|G^{1/2}(O^{-1}/(p\sqrt{w}) - a_* e_0^*) Q^{-1/2}\| \\ & \leq \sqrt{\frac{\lambda_{\max}(G)}{\lambda_{\min}(Q)}} \|O^{-1}/(p\sqrt{w}) - a_* e_0^*\|, \\ & a_* = \frac{\sqrt{3}}{8} (-i, 1, 0, 0)^\top. \end{aligned} \quad (\text{ES36})$$

Composition by the actual invertible conductor L has relative leading-operator error bounded by

$$\frac{\|O^{-1}/(p\sqrt{w}) - a_* e_0^*\| \|L^{-1}\|}{\|a_*\| \|e_0^* L^{-1}\|}. \quad (\text{ES37})$$

This denominator is part of the proved connecting map. A conductor approaching a directional resonance need not have a uniform bound here. Neither the growing RH kernel coefficient nor a complex-current phase has been assigned from the ES soft mode.

8.6. Verification. The exact checker derives ES19–20, all exterior first corrections, the complete rational inverse, integral families, and initial witness. Independent 110-digit SVD checks retain both core triples and test the four first singular corrections, soft energy, relative last-label component, and distinct-volume family. Large integer samples used for analytic asymptotics are not asserted to be primes. Source versions, reading coverage, and original/translated source distinctions are recorded in the accompanying ledger.

ACKNOWLEDGMENTS AND PROVENANCE

The Type I/II coordinate antecedents are due to Elsholtz and Tao [1, §2, Propositions 2.2 and 2.6]. The literal four-mark receiver and its determinant polynomial entered this project through the ES–Fable bridge [2]; the complete integral sign theorem and coefficient arrays used here are proved in [3]. The three-dimensional Jacobian construction that motivated that bridge was publicly announced by Levent Alpöge, with Akhil Mathew credited for the question and Claude Fable 5 for the work producing the construction; Tao’s later article is an independent exposition [4]. This paper proves only the ES receiver statements displayed here. Dirichlet’s theorem supplies infinitely many primes in the reduced class 13 mod 24 [5].

REFERENCES

- [1] Christian Elsholtz and Terence Tao, *Counting the number of solutions to the Erdős–Straus equation on unit fractions*, J. Aust. Math. Soc. 94 (2013), 50–105. arXiv:1107.1010v6; doi:10.1017/S1446788712000468.
- [2] The Clankers, *The odd receiving divisor pulled back to the Erdős–Straus locus*, 2026. Pinned proof, EZ37–EZ48.
- [3] The Clankers, *A global sign for the literal Erdős–Straus receiving determinant*, 2026. Pinned theorem and complete arithmetic certificates.
- [4] Terence Tao, *A digestion of the Jacobian conjecture counterexample*, What’s New, 21 July 2026. Article.
- [5] Peter Gustav Lejeune Dirichlet, *There are infinitely many prime numbers in all arithmetic progressions with first term and difference coprime*, translated by Ralf Stephan, arXiv:0808.1408 (2008); original work published 1837.